HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and deployed **optical network solutions** for data center connectivity, ensuring scalability across global infrastructure by integrating new hardware capabilities.
- Collaborated with hardware and software engineers to evaluate emerging **optical technologies**, contributing to architectural decisions for network infrastructure.
- Developed and automated **optical testing procedures** for qualifying optical modules, ensuring they met stringent performance and reliability standards.
- Partnered with network operations teams to identify and troubleshoot complex operational issues, implementing improvements to network quality and RMA processes.
- Created and maintained comprehensive **technical documentation** detailing network designs, test results, and operational procedures to enhance team knowledge sharing.
- Participated in **design reviews** for new network deployments, providing input on architecture and implementation plans to cross-functional teams.
- Managed production support for enterprise applications on **AWS**, including EC2 and S3, resolving P1/P2 incidents and ensuring SLA adherence.

SKILLS

Networking – Optical Network Design, Data Center Connectivity, Network Architecture, Routing Protocols, Network Scalability

Cloud Platforms – AWS, EC2, S3, CloudWatch

Operating Systems – Linux, Unix

Scripting & Automation – Python, Bash, Shell Scripting, Perl

Hardware & Software – Optical Hardware, Network Software, Vendor Collaboration

Development Practices – CI/CD, Jenkins, Git, Code Reviews, Agile/Scrum

PROJECTS

Data Center Interconnect Optical Network Design

- Designed a high-capacity optical network architecture to interconnect multiple data centers, focusing on **performance** and **scalability** for AWS infrastructure.
- Evaluated new **optical modules** and transceivers from various vendors, developing implementation plans for seamless integration into the existing network.
- Authored detailed **technical documentation** for the optical network design, including schematics, configuration guides, and testing protocols.

Automated Optical Module Qualification Framework

- Developed an automated testing framework using **Python** to qualify optical modules, reducing manual testing effort by approximately **40%**.
- Implemented data collection and analysis scripts to verify module performance against key metrics like bit error rate and power levels.
- Collaborated with hardware engineers to define qualification criteria and troubleshoot test failures, ensuring compliance with **AWS** standards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)